



from the labs



Computer Science and Artificial Intelligence Laboratory (CSAIL) at MIT is using robots to do their gardening

Developing a sixth sense

An Indian researcher wants you to develop a sixth sense that makes the entire world your personal computer. And his latest gizmo – made from off-the-shelf items and costing less than Rs 17,500 – might just do that!

Mihir Patkar

Pranav Mistry needs to make a phone call. While most of us would whip out our handset for this simple task, Pranav holds up his left palm in front of his chest. As if by magic, a number-pad appears on his fingers! His right index finger, tipped with a red cap, deftly dials out the digits and hits 'Call'. The earphones and mike – attached to the cellphone in his pocket – take over, and Pranav is happily yapping away.

Reality is a playtool for this 27-year-old. Studying for a PhD at the Massachusetts Institute of Technology's Media Lab, he already holds a Master of Arts and Science from the institute, in addition to a Master of Design from IIT Bombay.

"I'm a *designer*," Pranav laughs. "Exposure to diverse fields, from design and technology to art and psychology, gave me an interesting perspective to the world."

Born to an architect in Palanpur, a small town in northern Gujarat, Pranav has previously worked as a User Experience (UX) researcher at Microsoft India.

"I love to see technology from a design perspective, and vice versa. This vision reflects in almost all of my projects and research work," he says.

His new endeavour stands testament: called 'SixthSense', the tool sees what the user sees, and augments surfaces or objects he is interacting with. What's more, it's small and light enough to be

"We've evolved over millions of years to sense the world around us. When we encounter something, someone or some place, we use our five natural senses to perceive information about it; that information helps us make decisions and choose the right actions to take," he says. "But arguably, the most useful information that can help us make the right decision is not naturally perceivable with our five senses."

The data, information and knowledge that mankind has accumulated about everything is now increasingly available online.

"Although the miniaturisation of computing devices allows us to carry computers in our pockets, there is no link between our digital devices and our interactions with the physical world. Information is traditionally confined on paper or digitally on a screen," the engineer says.

Unveiling their project at the Technology, Entertainment, Design (TED) 2009 expo in February, Maes said, "I had been intrigued by this question whether we would evolve or develop a sixth sense, that would give us endless and easy access to meta-information that may be relevant to making a right decision."

That's where Maes' "genius" student came in. Pranav had previously worked in the field of augmented reality, in which real-world and computer-generated data are seamlessly blended into real footage in real-time.

"I've noticed that we humans are interested in information, not the computing. The idea of living in the real world, but still augmented with the advantages of the digital world, excites me," he says. "In fact, I don't like the idea of having two separate worlds. It should be seamless." So Pranav decided to push augmented reality one step further.

Turning sci-fi to reality

Steven Spielberg's 2002 science-fiction film *Minority Report* offered many glimpses into the future of technology. On the computing side, the audience gasped in wonder as Tom Cruise's character wore a special glove to manipulate a digital screen without touching it – operating it solely by gestures.

Spielberg had based this amazing interface on research pioneered by one of his science advisors – John Underkoffler. Since then, Underkoffler, an MIT alumnus, has developed this interface further at his company, Oblong. He recently demonstrated a working prototype of the same, called the 'g-speak'.

"Unfortunately, g-speak uses an array of 10 high-precision infra-red cameras, multiple projectors, and an expensive hardware setup," Pranav says, adding that it still served as one of his inspirations.

There are other such technologies too – although none as advanced as the g-speak – that share common hindrances – the



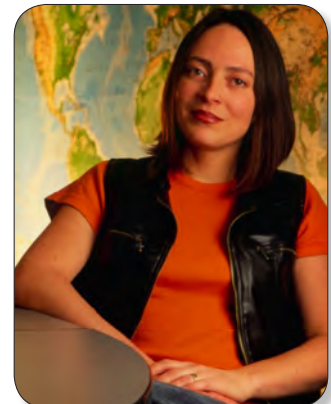
With a fully functional keypad projected on his palm, Mistry dials a number by tapping his fingers

worn around your neck like a pendant, or affixed to your cap.

"SixthSense attempts to free information from its confines by seamlessly integrating it with reality, and thus making the entire world your computer," he says. Pranav is associated on the project by the head of the MIT Media Lab's Fluid Interfaces Group, Pattie Maes, and fellow student Liyan Chang.

Creating a sixth sense

Multi-touch and gesture-based devices have been getting increasingly popular, Pranav says. Meanwhile, mobile devices are getting increasingly smaller, allowing us to be connected anywhere and all the time. So why not couple the two?



Pattie Maes, Head of MIT Media Lab's Fluid Interfaces Group